

Axel Lindberg

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EDUCATION

KTH Royal Institute of Technology

Stockholm, Sweden

MSc in Computer Science and Engineering

Aug. 2024 – May 2029

- **Relevant Coursework:** Artificial Intelligence, Software Engineering in Project Form, Probability Theory and Statistics with Applications to Machine Learning, Computer Organization and Components

EXPERIENCE

Software Project Lead – MetroPed

Dec. 2025 – June 2026

Embodied Social Agents Lab (ESAL), KTH – Industry Collaboration

Stockholm, Sweden

- Led a 6-developer team building a VR simulation of a Stockholm metro station for research on perceived safety, using Agile/Scrum to deliver a client-validated MVP ahead of schedule
- Managed sprint planning and project risks, keeping the team aligned with ESAL's research goals
- Engineered a custom impostor rendering pipeline in Unity and C# for crowd characters, significantly improving performance while preserving visual fidelity and enabling larger, more realistic crowds
- Designed the team's Git branching workflow and maintained comprehensive technical documentation, ensuring a scalable codebase and a seamless stakeholder handover

Production Operator (Blending & Bottling)

May 2024 – Aug. 2026

The Absolut Company

Åhus, Sweden

- Independently managed facility-wide blending and bottling operations across multiple high-volume production lines while maintaining strict regulatory compliance
- Partnered with automation and mechanical engineers to diagnose and resolve complex system failures, minimizing production downtime

PROJECTS

AI Calorie Tracker | *Python, FastAPI, PostgreSQL, OpenAI API, Docker, JavaScript*

Apr. 2026 – Present

- Built a full-stack LLM application that turns natural-language meal descriptions (typed or dictated via the Web Speech API) into calorie and macro data
- Used the OpenAI API with Pydantic structured outputs to parse free text into food items, quantities and units
- Grounded nutrition data in USDA FoodData Central, pre-filtering search results with RapidFuzz before an LLM selects or rejects the best match
- Implemented portion-to-gram conversion and nutrient scaling, with core calculation logic covered by unit tests
- Designed a FastAPI REST backend with a Dockerized PostgreSQL database and a lightweight Vanilla JS + Vite frontend

Bombberman (*Best Project Award*) | *C, RISC-V Assembly, CMake*

Nov. 2025 – Dec. 2025

- Recreated Bombberman in a 2-person team on a DE10-Lite FPGA board with a 32-bit RISC-V processor, awarded Best Project in KTH's IS1500 Computer Organization and Components
- Programmed core game systems in C and Assembly: player movement, bomb placement, explosions, collision detection and game state management
- Interfaced directly with hardware through memory-mapped I/O, delivering responsive real-time gameplay with no operating system

SKILLS

Languages: Python, Java, C, C#, JavaScript, SQL, HTML/CSS

AI & Backend: FastAPI, OpenAI API, Pydantic, PostgreSQL, RapidFuzz

Tools & Practices: Git, Docker, Unity, Claude Code, Agile/Scrum

Spoken Languages: Swedish (Native), English (Fluent)

Licenses: Swedish Driver's License (Körkort B)